

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.:

- **HTTP (HyperText Transfer Protocol):** The foundational protocol used to request and transmit web pages and data over the internet.
- **HTTPS (HyperText Transfer Protocol Secure):** The secure version of HTTP that encrypts data transferred between a web browser and a website to protect sensitive information.
- **FTP (File Transfer Protocol):** A standard protocol used to upload, download, and transfer computer files between a client and a server on a network.
- **SMTP (Simple Mail Transfer Protocol):** The standard communication protocol used to send and route email messages across the internet.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

- **IPv4** uses a 32-bit address format (e.g., 192.168.1.1), which limits it to roughly 4.3 billion unique IP addresses.
- **IPv6** uses a newer 128-bit alphanumeric format (e.g., 2001:0db8:85a3::8a2e:0370:7334), which provides a virtually limitless number of unique IP addresses to accommodate the growing number of devices.
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3. Run nslookup google.com on your system and write down the IP address you get.

(Note: The exact IP address will vary depending on your location and the specific Google server closest to you.)

If you run nslookup google.com in your command prompt or terminal, you will typically get both an IPv4 and an IPv6 address. A common result looks like this:

- **IPv4 Address:** 142.250.190.46
- **IPv6 Address:** 2607:f8b0:4005:80b::200e

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

- **200 (OK):** The request was successful, and the server delivered the requested webpage or data.
- **301 (Moved Permanently):** The requested resource has been permanently moved to a new URL, and the browser will automatically redirect you.
- **404 (Not Found):** The server could not find the requested webpage, usually because the URL is incorrect or the page was deleted.

- **500 (Internal Server Error):** The server encountered an unexpected error on its end and could not process your request.
- **403 (Forbidden):** The server understood your request, but you do not have the necessary permissions or authorization to view the page.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with `www.google.com` as an example.

- **Root (.):** This is the very top of the DNS hierarchy. While you don't type it in your browser, fully qualified domain names technically end in a silent dot (e.g., `[www.google.com](https://www.google.com).`).
- **TLD (Top-Level Domain):** This is the suffix at the end of the domain name. In our example, **.com** is the TLD, which stands for commercial.
- **Domain:** This is the primary, registered name of the website. In our example, **google** is the domain name.
- **Subdomain:** This is the prefix added before the domain to direct traffic to a specific part of the site. In our example, **www** (World Wide Web) is the subdomain.